

NEAR-MISS INCIDENT REPORT

Report No. : NMR-2019-047
Date of Incident: 14 March 2019
Time of Incident: 14:32 IST
Location : Bay 4, Zone C ? Furnace Pre-heat Section
Equipment Tag : GB-14 (Draeger Polytron 2 XP Combustible Gas Detector)
Report Prepared : Priya Nair, HSE Manager
Reviewed By : Mohan Rao, Plant Manager
Date Submitted : 15 March 2019

1. INCIDENT DESCRIPTION

At 14:32 on 14 March 2019, equipment tag GB-14 (combustible gas detector installed at Bay 4, Zone C, at elevation 3.2m) registered a sustained alarm reading of 27% LEL (Lower Explosive Limit). The reading had been gradually rising from 11% LEL since 12:15 IST but was not escalated by the shift operator (Suresh Kumar, Operator-II) until it crossed the 25% LEL threshold trigger.

At the time of the alarm, a hot work permit (Permit No. HWP-2019-112) had been issued for gas cutting operations being carried out by contractor technician Balraj Tiwari (Sarwa Contractors Pvt. Ltd.) approximately 8 metres from the GB-14 sensor location. The permit had been authorised by Rajiv Sharma (Maintenance Supervisor) at 13:45 IST.

On receiving the alarm escalation, HSE Officer Priya Nair immediately issued a STOP WORK order. All hot work was halted. The area was ventilated using forced-draft blowers for 45 minutes. A second gas reading at 15:20 showed 4% LEL. Normal operations resumed at 16:10.

No fire, explosion, or injury occurred. This event is classified as a NEAR-MISS ? Category 2 (High Potential).

2. IMMEDIATE CAUSES

- (a) GB-14 sensor had not been calibrated since 22 September 2018 (last calibration record: Work Order WO-3591, Technician: Anil Mehta). The manufacturer (Draeger) recommends 6-monthly calibration. At the time of this incident the sensor was 174 days overdue.
- (b) Operator Suresh Kumar did not escalate the rising gas reading (11%?27% LEL over 137 minutes) per the Shift Handover Procedure SHP-04 which requires supervisor notification at 15% LEL.
- (c) Hot Work Permit HWP-2019-112 was issued without a pre-work gas reading check from GB-14, contrary to OISD-STD-105 Clause 4.3.1 which mandates a fresh gas reading within 30 minutes of hot work commencement.

3. ROOT CAUSES

- (a) No automated system exists to block hot work permit issuance when nearby gas sensor readings exceed 10% LEL. Permits are currently issued on paper without cross-checking live sensor data.
- (b) Calibration tracking is manual (maintained in a paper logbook in the Maintenance Supervisor's office). No digital reminder or work order auto-generation exists when calibration is overdue.
- (c) Operator training records show Suresh Kumar last attended gas hazard escalation training in February 2017 ? over 2 years prior to this incident.

4. CORRECTIVE ACTIONS ISSUED

- CA-01 | GB-14 to be calibrated within 72 hours. (Assigned: Anil Mehta. Due: 17 Mar 2019)
- CA-02 | All hot work permits in Bay 4 to require a live gas reading sign-off from the nearest sensor (GB-14 or GB-17) within 30 minutes of permit issuance. (Immediate)
- CA-03 | Suresh Kumar to attend refresher training ? Gas Hazard Response, Module 3.